Resource-Aware Staged Feature Acquisition for Heart Disease Triage: A Multi-Site Safety and Generalization Audit

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***Abstract*—Most machine learning models for heart disease prediction are evaluated as if all diagnostic features are available at the moment of patient contact. This assumption works well for benchmarking, but it ignores real-world triage constraints. In practice, intake variables are readily available, whereas downstream diagnostic tests require additional time, cost, and infrastructure. Furthermore, the literature generally lacks staged safety audits that incorporate bypass accounting, threshold sweeps, and site-held-out validation. We present a retrospective staged-acquisition workflow simulation using a two-stage routing framework, evaluated on a heterogeneous 920-record multi-site UCI heart disease dataset. This framework separates readily available intake variables (Tier 1) from a comprehensive downstream diagnostic profile (Tier 2). Under 5-fold cross-validation, the cascade avoided 25.1% of downstream diagnostic evaluations—meaningfully reducing unnecessary testing—while maintaining an accuracy of 84.5%. This matched the full-feature Tier 2 model (84.3% accuracy, 0.904 AUROC) within fold-level variability. However, our evaluation also exposed significant generalization risks. Leave-one-site-out testing dropped the pooled cascade accuracy to 78.0%. Held-out safety audits further revealed that an efficiency-driven routing gate bypassed 4 true-positive cases out of 45 bypassed patients, whereas a stricter threshold safely avoided 9.8% of downstream tests with zero bypassed positives. This study demonstrates that while staged acquisition can systematically reduce unnecessary downstream diagnostic testing, the associated safety and generalization trade-offs must be explicitly quantified, offering a practical framework for auditing clinical algorithms before deployment.**

***Index Terms*—clinical decision support, heart disease prediction, staged feature acquisition, selective classification, calibration, decision curve analysis, SHAP**

I. Introduction

Cardiovascular disease remains one of the leading causes of morbidity and mortality worldwide, making accurate clinical decision support an important objective for machine-learning research. Consequently, cardiovascular prediction has long served as a benchmark problem for statistical and machine-learning models [1], [2]. However, this benchmarking tradition exists within a broader clinical-AI literature arguing that predictive systems should be evaluated in the context of their intended clinical workflow, rather than solely by aggregate discrimination on convenient retrospective splits [3]–[5]. A major operational mismatch in current benchmarking is the implicit assumption that all diagnostic variables are simultaneously available at the moment of patient presentation.

In routine clinical practice, however, triage is inherently sequential. Readily available intake variables—including age, sex, chest pain characteristics, and resting blood pressure—are collected during the initial clinical encounter. In contrast, downstream diagnostic variables such as thallium stress-test categories, major vessel counts, electrocardiographic findings, and other specialized investigations require additional clinical resources, equipment, personnel, and time before becoming available. Consequently, although full-feature prediction models may achieve strong statistical discrimination, they may not accurately reflect the operational realities of early-stage clinical decision making.

Prior cardiovascular machine-learning studies have demonstrated strong predictive performance when complete downstream diagnostic profiles are available [6]–[8]. At the same time, algorithmic paradigms such as selective classification, reject-option learning, budgeted feature acquisition, and learning-to-defer provide theoretical frameworks for optimizing feature acquisition or routing decisions under uncertainty [26]–[30]. However, to our knowledge, comparatively limited attention has been devoted to evaluating these staged workflows from the perspective of clinically interpretable safety auditing. In particular, relatively few studies jointly examine explicit bypass safety accounting, threshold sensitivity analysis, calibration, and leave-one-site-out (LOSO) validation within a staged cardiovascular workflow. Consequently, an important gap remains in understanding how staged diagnostic acquisition influences both cross-site generalization and clinically relevant safety-efficiency trade-offs.

To address this gap, we hypothesize that a staged routing workflow can reduce downstream diagnostic feature acquisition while preserving predictive performance under explicit, quantifiable safety constraints. Rather than proposing a novel optimization algorithm, we present a retrospective workflow-auditing methodology. Specifically, we simulate a fixed, clinically interpretable two-tier routing system that separates immediately available intake variables from downstream diagnostic features. We intentionally employ a fixed routing policy instead of a learned acquisition or learning-to-defer strategy to prioritize workflow interpretability, reproducibility, and explicit safety auditing over algorithmic optimization. Model transparency is further supported through supplementary SHAP (SHapley Additive exPlanations) analyses [33], which are used solely to provide local feature attribution for Tier 2 predictions rather than to establish causal explanations or validated clinical interpretability [34].

This paper makes the following contributions:

- We evaluate a retrospective two-tier cardiac triage simulation to quantify the potential reduction in downstream diagnostic feature acquisition obtained by separating readily available intake variables from delayed diagnostic features.
- We analyze the safety-efficiency trade-offs between conservative and uncertainty-band routing strategies using Tier 1-only and full-feature baselines as reference models.
- We report cross-validation, held-out, and leave-one-site-out (LOSO) evaluations to distinguish favorable random-split performance from realistic cross-site generalization behavior.
- We comprehensively audit the routing mechanism using explicit bypass safety accounting, probability calibration, threshold sensitivity analysis, and decision-curve analysis to characterize the safety implications of fixed staged routing.
- We provide exploratory subgroup analyses and supplementary SHAP-based local feature attributions as transparency mechanisms supporting workflow traceability.

Overall, this work contributes a reproducible workflow simulation and safety-auditing methodology for evaluating staged clinical feature acquisition. Rather than introducing a new machine-learning algorithm, calibrated clinical risk score, validated explanation system, or deployment-ready medical device, the study provides a practical evaluation framework for systematically assessing the safety and generalization characteristics of staged clinical decision-support workflows prior to deployment.

Staged triage workflow evaluated in this study. The implemented application bypasses Tier 2 only for low-risk Tier 1 cases. The uncertainty-band policy, which also bypasses high-risk Tier 1 cases, is evaluated only as a cost-first ablation. SHAP output is limited to ranked local attribution factors for Tier 2 decisions.

II. Related Work

A. Cardiovascular Prediction and Clinical Workflows

Machine-learning models have been widely applied to cardiovascular risk prediction using routine clinical records, imaging, and public benchmark datasets [6]–[8]. Within clinical cardiology, however, diagnostic decision-making is inherently sequential, with patients typically undergoing progressively more expensive or invasive investigations based on earlier clinical findings. While several cardiovascular prediction studies report strong discrimination using complete diagnostic feature sets, comparatively fewer studies explicitly evaluate how such models align with staged clinical workflows. Furthermore, the external validity of cardiovascular prediction models is frequently affected by cohort composition, informative missingness, and site-specific data-generating processes [5], [13], [14]. To better characterize these generalization challenges, the present study incorporates both conventional random-split evaluation and leave-one-site-out (LOSO) validation.

B. Staged Acquisition, Selective Prediction, and Learning-to-Defer

The broader machine-learning literature has extensively investigated feature acquisition under resource constraints through cost-sensitive learning [23], [24], cascade classifiers [25], selective classification [26]–[28], and learning-to-defer frameworks [29], [30]. These approaches provide principled mechanisms for feature acquisition, abstention, or routing decisions under uncertainty. However, to our knowledge, comparatively limited attention has been devoted to evaluating these routing strategies from the perspective of clinically interpretable workflow auditing. In particular, relatively few studies jointly assess staged diagnostic workflows using explicit bypass safety accounting, threshold sensitivity analysis, calibration auditing, and cross-site validation. Rather than introducing a new routing algorithm, the present work evaluates a fixed, clinically interpretable staged workflow in order to systematically audit its safety and operational characteristics.

C. Clinical Evaluation, Calibration, and Decision Support

Accuracy alone provides an incomplete assessment of clinical prediction models, particularly under class imbalance or dataset shift. Complementary metrics including AUROC, sensitivity, specificity, balanced accuracy, Matthews correlation coefficient (MCC), probability calibration, and decision-curve analysis (DCA) evaluate different aspects of model reliability and clinical usefulness [10], [15]–[18]. When prediction probabilities directly influence staged routing decisions, calibration quality and threshold-dependent utility become especially important. To our knowledge, comparatively limited attention has been devoted to applying calibration auditing, decision-curve analysis, and continuous threshold sensitivity analysis as integrated safety evaluations for staged diagnostic workflows. Accordingly, these analyses are employed in the present study as post-hoc workflow audits rather than evidence of deployment readiness.

D. Model Traceability, Missingness, and Subgroup Analysis

Healthcare datasets frequently contain informative missingness arising from institutional workflows rather than purely random absence [31]. Likewise, aggregate performance metrics may obscure clinically important differences across demographic subgroups or acquisition sites [35]. Explainable-AI methods such as SHAP provide principled additive feature attributions for individual predictions [33]. However, SHAP alone does not establish causal relationships, fairness, or clinical usefulness [34]. Consequently, the present study treats missingness analysis, subgroup evaluation, and SHAP-based feature attribution as complementary transparency mechanisms that support workflow traceability for Tier 2 predictions rather than as standalone evidence of clinical validity.

TABLE I
COMPARISON OF REPRESENTATIVE PREDICTION FRAMEWORKS AND EVALUATION METHODOLOGIES

Framework / Literature	Staged Routing	Primary Objective	External / LOSO Validation	Calibration / DCA	Explicit Bypass Audit	Interpretability
Standard Full-Feature Prediction [6], [8]	No	Predictive Performance	Variable	Variable	Not Applicable	Variable
Budgeted Feature Acquisition [23], [24]	Yes	Acquisition Cost Optimization	Limited	Limited	Limited	Limited
Selective Classification / Learning-to-Defer [28]–[30]	Yes	Decision Risk Optimization	Variable	Variable	Not Primary Focus	Limited
Proposed Workflow Audit	**Yes (Fixed)**	**Clinical Workflow Safety and Efficiency**	**Yes (LOSO)**	**Yes**	**Yes (Explicit)**	**SHAP-Based Transparency**

E. Research Gap Summary

Collectively, prior work has addressed feature acquisition cost, selective prediction, calibration, interpretability, or clinical evaluation as largely independent problems. To our knowledge, comparatively limited attention has been devoted to comprehensive evaluations of staged cardiovascular workflows that jointly incorporate explicit bypass safety accounting, probability calibration, threshold sensitivity analysis, decision-curve analysis, and cross-site validation within a single evaluation framework. Consequently, it remains unclear whether staged diagnostic acquisition can preserve predictive performance while maintaining acceptable safety characteristics across heterogeneous clinical settings. The present work addresses this gap through a reproducible workflow-auditing methodology that emphasizes safety, generalization, and operational transparency rather than algorithmic novelty.

III. METHODS

A. Study Design

This study retrospectively simulated staged clinical decision-making using previously collected diagnostic records from a publicly available, fully de-identified dataset. No patient interaction occurred, no identifiable personal information was collected, and no intervention was performed. The study follows the reporting principles of TRIPOD and TRIPOD+AI for prediction-model studies [11], [12]; potential sources of bias are considered using the principles of PROBAST and clinical-AI validation guidance [5], [19]. Although CONSORT-AI, SPIRIT-AI, and DECIDE-AI provide guidance for prospective clinical AI evaluation, the present work did not involve a clinical trial or deployment study [20]–[22]. These guidelines informed the reporting structure but were not formally applied as checklist compliance instruments.

Four operating modes were evaluated:

1) **Tier 1 only:** an intake model using immediately available triage variables.
2) **Tier 2 full profile:** a diagnostic model using all processed clinical features.
3) **Conservative cascade:** the primary workflow policy, where low-risk Tier 1 predictions bypass Tier 2 and all remaining cases proceed to Tier 2.
4) **Uncertainty-band policy:** a cost-first ablation in which low-risk and high-risk Tier 1 predictions bypass Tier 2, while only intermediate-risk cases receive downstream evaluation.

Model generalization was primarily assessed using stratified five-fold cross-validation and leave-one-site-out (LOSO) validation. LOSO evaluation provides a realistic assessment of institutional generalization across sites with differing disease prevalences and acquisition protocols. A stratified 80/20 split was additionally reserved as an independent held-out development set for post-hoc confusion-matrix analysis, calibration diagnostics, threshold sensitivity analysis, bypass-safety accounting, and decision-curve analysis.

B. Dataset and Eligibility

The dataset was obtained from the publicly available UCI Heart Disease repository and comprises records collected from four participating institutions: Cleveland, Hungary, VA Long Beach, and Switzerland. Institutional Review Board (IRB) approval was not required because all records are publicly available and fully de-identified.

The processed dataset contained 920 patient records comprising 13 raw clinical predictor variables and one target variable. Unlike several benchmark studies that evaluate only the filtered Cleveland subset [8], the present study utilizes the complete multi-site cohort to enable assessment of cross-site generalization under heterogeneous clinical distributions.

The binary prediction target was defined as

$$y = \begin{cases} 1, & \text{target} > 0, \\ 0, & \text{target} = 0. \end{cases} \tag{1}$$

This formulation produced 509 positive and 411 negative cases. No patient records were excluded prior to the train–test split in order to avoid potential bias introduced by complete-case filtering. Data cleaning consisted of screening continuous variables for physiologically implausible values; only resting cholesterol measurements equal to zero ($n = 172$) were considered invalid and converted to missing values prior to imputation.

Substantial missingness was present in several variables, particularly the number of major vessels (`ca`), ST depression induced by exercise (`oldpeak`), and multiple categorical diagnostic features. Detailed preprocessing procedures, including feature encoding, missing-value imputation, and normalization, are described in the subsequent Preprocessing subsection.

A stratified 80/20 split produced 736 training records and 184 held-out records. Exact reproducibility details, including the random seed, software environment, and library versions, are provided in the Code Availability section. Table II summarizes the dataset composition and disease prevalence across participating sites.

C. Preprocessing

To prevent data leakage, all transformations were implemented using `scikit-learn Pipeline` and

TABLE II
CHARACTERISTICS OF THE UCI HEART DISEASE COHORT

Source Site or Split	Records	Positive	Negative	Prevalence
Cleveland	304	139	165	0.457
Hungary	293	106	187	0.362
VA Long Beach	200	149	51	0.745
Switzerland	123	115	8	0.935
Full Dataset	**920**	**509**	**411**	**0.553**
Held-Out Development Set	**184**	**102**	**82**	**0.554**

`ColumnTransformer` objects. Every imputer, encoder, and scaler was fit strictly on the training partition within each split. The preprocessing workflow followed a fixed order: data cleaning, missingness flagging, imputation, encoding, and scaling.

Cholesterol values of zero were treated as physiologically invalid and converted to missing values before continuous imputation; 172 of 920 records had cholesterol equal to zero in the dataset. Because missingness in healthcare data may be informative [31], explicit binary missingness indicators were generated for clinically meaningful absence patterns: `ca_missing` (611 missing), `chol_missing_or_zero` (172 missing or zero), and `oldpeak_missing` (62 missing).

Continuous variables were imputed using iterative chained imputation (MICE, `max_iter=15`) fitted on the training split. This approach was selected because it can better preserve multivariate relationships among clinical variables than simple univariate imputation strategies. Missing categorical values were imputed with a distinct sentinel value of -1 before one-hot encoding. This avoids collision with genuine positive binary values such as fasting blood sugar $= 1$ or exercise-induced angina $= 1$ while preserving numeric consistency in the pipeline. Categorical features were processed using one-hot encoding with `handle_unknown='ignore'`. Finally, continuous variables were scaled using Min-Max normalization to a $[0, 1]$ range. This preserves comparable feature scales for the K-Nearest Neighbors component and supports stable numerical optimization for Logistic Regression in the Tier 2 ensemble.

The final encoded Tier 2 matrix contained 37 features.

D. Feature Tiers

Features were partitioned according to their real-world clinical availability. Tier 1 was restricted to readily available intake variables:

$$X_1 = \{\text{age}, \text{sex}, \text{chest pain type}, \text{resting blood pressure}\}. \tag{2}$$

After one-hot encoding, Tier 1 contained exactly eight columns: age (1), sex indicators (2), chest-pain indicators (4), and resting blood pressure (1).

Tier 2 used the complete encoded diagnostic profile:

$$X_2 = X_1 \cup X_{\text{diagnostic}}, \tag{3}$$

where $X_{\text{diagnostic}}$ includes cholesterol, fasting blood sugar, resting electrocardiogram, maximum heart rate, exercise-induced angina, ST depression, slope, major vessels, thalassemia or stress-test category, and the explicitly generated missingness indicators. The final encoded Tier 2 matrix contained 37 features.

Several Tier 2 features, especially major-vessel counts and thalassemia/stress-test variables, are downstream diagnostic measurements. Their downstream nature is the primary reason they are staged rather than treated as immediately available intake information.

E. Model Architecture

The objective of model development was to construct stable predictive components for evaluating staged clinical workflows rather than maximizing benchmark performance through extensive hyperparameter optimization.

The Tier 1 gatekeeper was implemented as a Random Forest classifier with 600 estimators, maximum tree depth of 4, minimum samples split of 2, and random state 43 [32]. The Tier 2 model was implemented as an equally weighted soft-voting ensemble (`VotingClassifier`) comprising two Random Forest classifiers, Logistic Regression, and K-Nearest Neighbors (KNN). The first Tier 2 Random Forest used 200 estimators, maximum depth 12, minimum samples split 10, and random state 7, while the second used 1000 estimators, maximum depth 5, minimum samples split 10, and random state 43. Logistic Regression was configured with 5000 maximum iterations to ensure convergence, and the KNN classifier used seven nearest neighbors.

Hyperparameters were selected through manual empirical experimentation during prototype development and fixed prior to the final evaluation. No nested hyperparameter optimization is claimed. Consequently, the reported results should not be interpreted as the maximum achievable predictive performance, but rather as the performance of a fixed workflow evaluated under reproducible conditions.

Each model maps its respective encoded feature space to a positive-class probability:

$$f_1 : \mathbb{R}^8 \to [0, 1], \qquad p_1 = f_1(X_1), \tag{4}$$

$$f_2 : \mathbb{R}^{37} \to [0, 1], \qquad p_2 = f_2(X_2), \tag{5}$$

where $\mathbb{R}^8$ and $\mathbb{R}^{37}$ denote the encoded Tier 1 and Tier 2 feature spaces, respectively. For the Tier 2 ensemble, the final prediction probability is obtained by equal-weight soft voting:

$$p_2 = \frac{1}{4}\sum_{i=1}^{4} p_{2,i}, \tag{6}$$

where $p_{2,i}$ denotes the positive-class probability predicted by the i^{th} base classifier.

F. Routing Policies

The staged workflow employed lower and upper routing thresholds,

$$\theta_L = 0.30, \qquad \theta_H = 0.70. \tag{7}$$

The primary workflow evaluated throughout the study was the conservative routing policy,

$$r_{\text{app}}(p_1) = \begin{cases} \text{stop after Tier 1}, & p_1 \leq \theta_L, \\ \text{request Tier 2}, & p_1 > \theta_L, \end{cases} \tag{8}$$

where patients predicted to be low risk bypass downstream feature acquisition, while all remaining patients undergo Tier 2 evaluation.

For comparison, a cost-first uncertainty-band policy was evaluated as an ablation experiment:

$$r_{\text{band}}(p_1) = \begin{cases} \text{low-risk bypass}, & p_1 \leq \theta_L, \\ \text{request Tier 2}, & \theta_L < p_1 < \theta_H, \\ \text{high-risk bypass}, & p_1 \geq \theta_H. \end{cases} \tag{9}$$

The routing thresholds were selected empirically during prototype development as operational workflow baselines rather than calibrated clinical decision thresholds. Consequently, post-hoc threshold sensitivity analyses were performed to evaluate the robustness of workflow performance under alternative routing thresholds.

G. SHAP Attribution Layer

For patients routed to Tier 2, local feature attributions were computed using the SHAP (SHapley Additive exPlanations) Permutation Explainer. A model-agnostic explainer was selected because the Tier 2 predictor is a heterogeneous soft-voting ensemble rather than a single tree-based model.

Feature importance was quantified using the absolute SHAP value,

$$\text{Impact}_i(x) = |\phi_i(x)|, \tag{10}$$

where $\phi_i(x)$ denotes the local contribution of feature i for patient x.

The interface reports the highest risk-raising and risk-lowering features together with their attribution direction. These outputs serve exclusively as an inspectable model trace. The study does not claim that SHAP attributions represent causal mechanisms, patient-facing explanations, or validated clinician decision aids.

H. Metrics and Statistical Analysis

Because the primary objective of this study is workflow evaluation rather than algorithmic optimization, the principal evaluation endpoints were Tier 2 profile avoidance, bypass safety (quantified by bypassed positive cases), and cross-site generalization using leave-one-site-out (LOSO) validation.

Standard classification metrics—including accuracy, balanced accuracy, precision, recall, specificity, F1-score, Matthews Correlation Coefficient (MCC), and confusion matrices—were reported as complementary discriminative measures. MCC was included because it remains informative under moderate class imbalance. Cross-validation results are reported as fold-wise means with corresponding standard deviations.

TABLE III
FIVE-FOLD CROSS-VALIDATION SUMMARY

Model or policy	Accuracy	AUROC	Tier 2 profile avoidance
Tier 1 only	0.777 ± 0.027	0.843 ± 0.036	–
Tier 2 full profile	0.843 ± 0.018	0.904 ± 0.035	0.0%
Conservative cascade	0.845 ± 0.022	–	$25.1\% \pm 3.3\%$
Uncertainty-band policy	0.826 ± 0.026	–	$72.4\% \pm 2.5\%$

Area Under the Receiver Operating Characteristic (AUROC) and Brier score are reported only for the stand-alone Tier 1 and Tier 2 probability models. These metrics were not used to rank staged routing policies because cascaded workflows combine predictions from multiple decision stages and therefore do not produce a single uniformly calibrated probability distribution.

Tier 2 profile avoidance was quantified as

$$A_{T2} = \frac{N - N_{T2}}{N}, \tag{11}$$

where N denotes the total number of evaluated patients and N_{T2} represents the number routed to Tier 2. Larger values of A_{T2} indicate greater downstream diagnostic profile avoidance. This metric reflects workflow efficiency rather than monetary cost.

Decision-curve analysis (DCA) was performed across threshold probabilities ranging from 0.05 to 0.60. Net benefit was computed as

$$\text{NB}(p_t) = \frac{\text{TP}}{N} - \frac{\text{FP}}{N} \cdot \frac{p_t}{1 - p_t}, \tag{12}$$

where p_t denotes the decision threshold [18]. For staged policies, the final probability available at the terminal decision point (p_1 for bypassed patients and p_2 for Tier 2 patients) was used when computing decision curves.

Calibration performance was assessed using Brier score, 10-bin Expected Calibration Error (ECE), and equal-width reliability diagrams. Finally, bypass safety was quantified by recording the number of true-positive patients incorrectly assigned to the low-risk bypass branch. Within the context of staged clinical triage, any bypassed positive case was treated as a workflow safety failure.

IV. RESULTS

A. Cross-Validation and Site-Held-Out Generalization

Stratified 5-fold cross-validation produced more robust and stable estimates than the held-out development split. Table III shows that the conservative cascade reached 0.845 ± 0.022 mean accuracy while achieving $25.1\% \pm 3.3\%$ Tier 2 profile avoidance at the original 0.30 low-risk threshold. The full Tier 2 model reached 0.843 ± 0.018 accuracy and 0.904 ± 0.035 AUROC. The observed difference between the conservative cascade and the full Tier 2 model was smaller than the observed fold-to-fold variability. Since the study was designed as a workflow audit rather than an algorithm comparison, no formal hypothesis testing was performed.

Leave-one-site-out (LOSO) validation provided a more stringent generalization evaluation. Table IV shows a pooled

TABLE IV
LEAVE-ONE-SITE-OUT CONSERVATIVE CASCADE

Held-out site	N	Prev.	Maj.	Accuracy	Balanced Acc.	MCC	Sens.	Spec.
Cleveland	304	0.457	0.543	0.783	0.786	0.570	0.820	0.752
Hungary	293	0.362	0.638	0.799	0.820	0.615	0.896	0.743
Switzerland	123	0.935	0.935	0.813	0.726	0.276	0.826	0.625
VA Long Beach	200	0.745	0.745	0.730	0.683	0.344	0.779	0.588
Pooled	920	0.553	0.553	0.780	0.775	0.554	0.825	0.725

conservative-cascade accuracy of 0.780, balanced accuracy of 0.775, MCC of 0.554, sensitivity of 0.825, and specificity of 0.725. Switzerland and VA Long Beach had high disease prevalence, and the cascade performed below the corresponding majority-class accuracy baseline because of the extreme prevalence imbalance at both sites. This demonstrates that simple classification accuracy can be misleading under severe prevalence shift. Because LOSO evaluates one distinct, non-overlapping split per site, no fold-wise standard deviations are reported. The substantial degradation under LOSO evaluation compared with random-split validation highlights the importance of explicit cross-site validation.

B. Held-Out Development Split

Table V summarizes the 184-record held-out development split. The full Tier 2 model achieved 90.2% accuracy, 0.895 balanced accuracy, 0.805 MCC, and 0.920 AUROC. The conservative cascade achieved 89.7% accuracy, 0.890 balanced accuracy, and 0.793 MCC while achieving 24.5% Tier 2 profile avoidance. The uncertainty-band policy achieved 86.4% accuracy with 66.8% Tier 2 profile avoidance. The higher held-out accuracy reflects sampling variability in a single split and should not be interpreted as superior evidence to the cross-validation or LOSO analyses. These held-out point estimates are reported only as supplementary descriptive evidence; LOSO remains the primary generalization estimate.

The corresponding confusion matrices were: Tier 1, TN=58, FP=24, FN=15, TP=87; Tier 2, TN=68, FP=14, FN=4, TP=98; conservative cascade, TN=68, FP=14, FN=5, TP=97; and uncertainty-band policy, TN=62, FP=20, FN=5, TP=97. An exact two-sided McNemar test between Tier 2 and the conservative cascade produced $p = 1.000$, but this comparison had only one discordant pair, and therefore statistical power was extremely limited.

C. Bypass Safety and Threshold Sensitivity

The implemented 0.30 low-risk threshold routed 45 held-out patients to the bypass branch. Four of those 45 patients were positive cases, yielding a bypass NPV of 91.1%. This safety finding does not support a hypothetical discharge-style workflow at the 0.30 threshold. The threshold sweep in Table VI demonstrates why Tier 2 profile avoidance and bypass safety must be analyzed together. On this split, a 0.15 threshold achieved 9.8% Tier 2 profile avoidance (18 of 184 profiles) and recorded no bypassed positive cases. At 0.20, the first bypassed positive appeared.

Thus, the 25% Tier 2 profile avoidance result is best interpreted as an efficiency-oriented operational baseline, not as a

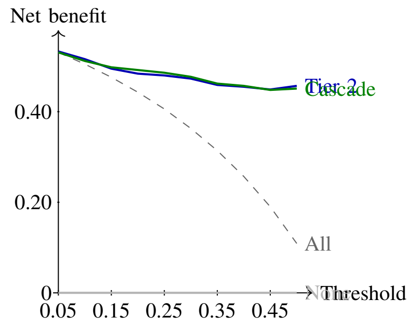

Fig. 1. Held-out decision-curve audit for the full Tier 2 model and conservative cascade. Net benefit is shown without confidence bands and should be interpreted as development-split evidence only.

clinically safe rule-out threshold. If zero bypassed positives are required, the empirically observed threshold on the held-out development split is 0.15, with substantially lower Tier 2 profile avoidance. These observations illustrate the inherent trade-off between diagnostic efficiency and patient safety. It is an observed development-split threshold, not a validated clinical cutoff.

D. Decision-Curve and Calibration Audits

Fig. 1 shows held-out decision-curve results for the full Tier 2 model and conservative cascade. Both models had higher net benefit than treat-all and treat-none strategies across the plotted threshold range. The cascade was close to the full Tier 2 model at several thresholds, but no confidence bands were estimated via bootstrap resampling; the figure is therefore a descriptive assessment of apparent clinical utility on the held-out split rather than a definitive clinical utility claim.

Table VII summarizes the held-out probability audit. Raw Tier 1 probabilities produced a Brier score of 0.153, AUROC of 0.844, and an Expected Calibration Error (ECE-10) of 0.052. Although the Tier 2 ensemble achieved superior discrimination (AUROC 0.920), it exhibited a higher ECE than Tier 1, indicating that improved discrimination does not necessarily imply improved calibration. These properties evaluate different aspects of probabilistic prediction. Five-fold isotonic calibration fitted on the training split slightly reduced ECE to 0.048 but worsened the Brier score and AUROC. The calibrated model was therefore not substituted into the deployed pipeline or reported cascade metrics. These results support the paper's conservative interpretation: the 0.30 and 0.70 gates are prototype routing thresholds, not calibrated clinical risk cutoffs.

E. Exploratory Subgroup Audit

Table VIII reports held-out conservative-cascade performance by sex and age band. The analysis is intentionally described as exploratory because the subgroup sample sizes are extremely small. For instance, while the female subgroup achieved 97.4% accuracy, this metric should be interpreted cautiously because of the limited sample size ($N = 38$).

TABLE V
HELD-OUT DEVELOPMENT PERFORMANCE

Model or policy	Accuracy	Balanced Acc.	Prec.	Sens.	Spec.	F1	MCC	AUROC	Brier
Tier 1 only	0.788	0.780	0.784	0.853	0.707	0.817	0.569	0.844	0.153
Tier 2 full profile	0.902	0.895	0.875	0.961	0.829	0.916	0.805	0.920	0.107
Conservative cascade	0.897	0.890	0.874	0.951	0.829	0.911	0.793	–	–
Uncertainty-band policy	0.864	0.854	0.829	0.951	0.756	0.886	0.730	–	–

TABLE VI
HELD-OUT LOW-THRESHOLD SWEEP FOR CONSERVATIVE ROUTING

θ_L	Tier 2 profile avoidance	Patients Bypassed	Bypassed Positives	Missed Positives
0.05	1.1%	2	0	0
0.10	4.9%	9	0	0
0.15	9.8%	18	0	0
0.20	15.8%	29	1	1
0.25	22.3%	41	3	3
0.30	24.5%	45	4	4
0.40	30.4%	56	8	8
0.50	39.7%	73	15	15
0.60	47.8%	88	24	15

TABLE VII
HELD-OUT PROBABILITY CALIBRATION AUDIT

Model	Brier	ECE-10	AUROC
Tier 1 raw	0.153	0.052	0.844
Tier 1 isotonic CV5	0.157	0.048	0.840
Tier 2 raw	0.107	0.097	0.920

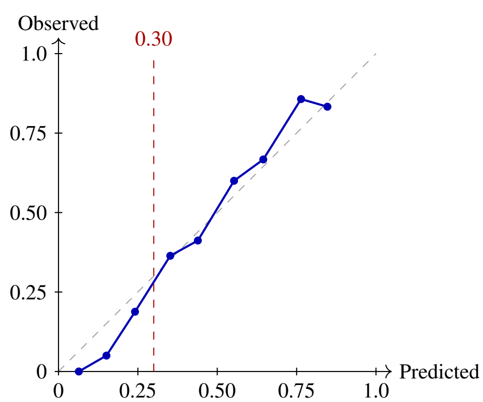

Fig. 2. Held-out reliability bins for raw Tier 1 probabilities. The 0.20–0.30 bin had an observed positive rate of 18.8%, illustrating why the 0.30 low-risk bypass branch cannot be considered clinically safe.

The oldest age band had only 20 records. These values are inadequate for establishing a formal fairness claim.

The lower specificity and MCC in older patients suggest that future validation should pre-specify age-stratified safety endpoints. No equality-of-odds, demographic parity, or subgroup safety claim is made.

TABLE VIII
HELD-OUT CONSERVATIVE CASCADE BY SUBGROUP

Subgroup	N	Prev.	Accuracy	Sens.	Spec.	MCC
Female	38	0.263	0.974	0.900	1.000	0.932
Male	146	0.630	0.877	0.957	0.741	0.734
$<$ 50 years	49	0.388	0.959	0.895	1.000	0.916
50–65 years	115	0.609	0.887	0.971	0.756	0.765
$>$ 65 years	20	0.650	0.800	0.923	0.571	0.545

F. Attribution Output

The application computes local Tier 2 attributions and displays the largest risk-raising and risk-lowering factors. In an illustrative held-out patient profile with a 31.7% final Tier 2 risk score, the highest-impact factors included major-vessel encoding, asymptomatic chest pain, thalassemia or stress-test category, ST-segment slope, and exercise-induced angina indicators. This example documents the structure of the attribution output only; it is not evidence that the displayed factors are causal or sufficient for autonomous clinical decision-making.

V. DISCUSSION

This study investigates staged feature acquisition as an alternative to conventional full-feature classification for heart disease prediction. Collectively, the results reveal a safety-efficiency frontier rather than a universally deployable routing threshold. Under cross-validation, the conservative cascade matched the full-profile model within fold-level variability while avoiding approximately one quarter of Tier 2 evaluations. However, the held-out development split exposed the underlying clinical risk: multiple positive patients appeared in the low-risk bypass branch at the initial 0.30 threshold. These findings also motivate greater attention to explicit diagnostic bypass safety, which is comparatively less emphasized than algorithmic routing performance in much of the selective prediction literature.

The threshold sweep further clarifies the practical interpretation of the resource savings. While a 0.30 threshold is operationally attractive (achieving 24.5% Tier 2 profile avoidance), it is unacceptable as a clinical rule-out mechanism within this audit. In contrast, the empirically observed 0.15 threshold on the held-out development split avoided only 9.8% of profiles but safely recorded zero bypassed positives. Therefore, the evidence supports the necessity of continuous safety-efficiency evaluation over the adoption of static, uncalibrated thresholds.

The substantial degradation observed during leave-one-site-out testing provides more critical generalization evidence than the favorable held-out point estimates. Pooled LOSO accuracy fell to 78.0%, and site-specific specificity was weak for high-prevalence cohorts like VA Long Beach and Switzerland. At these two sites, the cascade performed below the accuracy obtainable by always predicting the majority class. This behavior aligns with established literature on dataset shift in clinical AI; multi-site repositories inherently capture substantially different baseline prevalences and site-specific acquisition protocols [5], [14]. A staged workflow can therefore appear highly effective on randomized splits while failing to generalize robustly across distinct institutions.

Previous heart disease prediction studies have largely emphasized maximizing predictive discrimination using complete diagnostic feature sets. In contrast, the present study evaluates the operational consequences of staged feature acquisition, explicitly quantifying the trade-off between downstream diagnostic reduction, patient safety, and cross-site generalization. This complementary perspective extends conventional benchmarking by incorporating workflow-oriented evaluation criteria.

The calibration audit introduces further constraints on threshold interpretation. Tier 1 demonstrated lower calibration error (ECE) than the Tier 2 ensemble. The observed overconfidence in the Tier 2 ensemble suggests that higher discriminative performance does not necessarily translate into well-calibrated probabilities suitable for threshold-based routing. Because low-probability bins still contained positive cases, future clinical implementations must derive routing thresholds dynamically from cross-validated, utility-optimized calibration curves rather than pragmatic probability bands.

Finally, the interpretability component complements the routing audit rather than serving as its primary focus. SHAP attributions provide a traceable mechanism to rank local model drivers for escalated Tier 2 decisions. However, these attributions do not establish causal effects, clinical actionability, or improved clinician performance [34]. The appropriate interpretation is that the prototype can expose the factors most responsible for an individual model score; validating whether clinicians find those factor rankings reliable and useful requires a separate human-subjects evaluation.

A. Limitations

Several limitations prevent clinical deployment claims. First, the dataset is public, retrospective, temporally dated, and heterogeneous across source sites. Second, no prospective external cohort was evaluated to definitively prove generalization. Third, the evaluated routing thresholds were not derived from calibrated clinical risk, decision-curve optimization, or a pre-specified utility function. Fourth, the held-out bypass audit revealed multiple missed positive cases at the initially implemented threshold, meaning the current gate is unsafe for discharge decisions. Fifth, Tier 2 profile avoidance is strictly an operational proxy; no hospital cost model or time-motion study was performed. Sixth, no strong tabular learning baselines (e.g., gradient-boosted trees) or learned selective-prediction algorithms were evaluated, precluding claims of algorithmic superiority. Seventh, hyperparameters were fixed before the final evaluation and not optimized via nested cross-validation. Eighth, the subgroup audit was statistically underpowered and cannot formally establish algorithmic fairness. Ninth, certain Tier 2 diagnostic variables are obtained during later stages of the diagnostic pathway, which is acceptable for retrospective staging simulation but does not strictly model prospective causal diagnosis. Tenth, local SHAP attributions were not evaluated for clinician usability, robustness, or causal faithfulness. Finally, real-world clinical integration would require prospective validation, institutional governance, privacy controls, and regulatory assessment.

B. Reproducibility

The analysis utilized Python with `scikit-learn`, `pandas`, `NumPy`, `SHAP`, `joblib`, and `Streamlit`. The source data are publicly available from the UCI Heart Disease repository [1]. To ensure strict reproducibility, all random seeds were fixed. A standalone evaluation script regenerates all held-out metrics, bypass safety analyses, calibration summaries, reliability bins, threshold sweeps, decision-curve data, subgroup metrics, and site prevalence outputs.

C. Future Work

Future work should evaluate the routing gate using cross-validated calibration, nested hyperparameter optimization, utility-derived thresholds, and prospective external cohorts. Bypass safety should be reported across all cross-validation folds and leave-one-site-out settings, rather than strictly on a single held-out split. The fixed-gate architecture should be rigorously benchmarked against strong tabular learning baselines (e.g., gradient-boosted trees) and learned learning-to-defer policies. Furthermore, a realistic resource analysis should weight diagnostic variables by their site-specific monetary cost, waiting time, invasiveness, and staffing requirements. Finally, the SHAP-based traceability display requires a formal clinician-facing usability audit to determine if local factor rankings remain stable and useful within a live triage workflow.

VI. Conclusion

This study evaluated a staged feature-acquisition workflow for retrospective heart disease triage. The results demonstrate that reducing downstream diagnostic acquisition inevitably requires balancing efficiency against patient safety. On a 920-record multi-site cohort, the conservative cascade achieved $84.5\% \pm 2.2\%$ cross-validation accuracy while avoiding $25.1\% \pm 3.3\%$ of Tier 2 profiles. While held-out development metrics were strong, leave-one-site-out accuracy dropped to 78.0%, and the low-risk bypass branch contained missed positive cases at the prototype 0.30 threshold. A stricter, empirically observed 0.15 threshold on the held-out split yielded zero bypassed positives but avoided only 9.8% of profiles.

The collective evidence supports the feasibility of resource-aware staged clinical workflow evaluation and model traceability, but precludes immediate clinical deployment. Before such a framework could guide patient care, it would require robust probability calibration, principled threshold selection, prospective multi-site validation, rigorous cost modeling, and formal human-factors evaluation.

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